V18 E3B: A Breakthrough Strategy Beyond 0.921

Evidence-Complete Boundary Reranking for UIT DSC 2026 LegalIR under a strict 4B parameter cap

Deep research report and experimental decision plan

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1 Executive decision

Recommended strategy: do not broaden the candidate pool, do not begin with another generic Cross-Encoder (CE) fine-tune, and do not continue optimizing gates over the same V16 features. The highest-value next bet is **V18 E3B - Evidence-Complete Boundary Reranking**.

V18 keeps Stage-1, c35, BQB, the current Vietnamese CE, QREL-memory, and the safe slot-5 policy. It changes what evidence the CE sees. For each boundary candidate, it reconstructs legal structure, searches the full document for decisive articles and clauses, and packs selected non-redundant evidence into the CE's native 2304-token training window. It then compares challengers directly with the current tail anchor.

Four findings drive this decision:

1. **Most accessible headroom is already inside c35.** The locked c35 oracle is 0.967667 versus V16b at 0.9228333, leaving 0.0448337 absolute Recall@5 headroom without candidate expansion. P0 estimates that 67.06% of recoverable headroom is inside c35 [I3][I5].
2. **The CE is evidence-starved, not obviously capacity-starved.** V16b scores at most three passages per document at maximum length 768. The official model card says the deployed Vietnamese reranker was trained at sequence length 2304, with up to 2048 passage tokens [R3]. Recent long-document research consistently favors selecting and interacting over key blocks instead of feeding an arbitrary long prefix [R4][R5][R6].
3. **Prior failures mostly rearranged old evidence.** GateCal, Prototype-v2, V17 setwise ranking, c50, prefix512, generic hard-negative LoRA, frozen ColBERT, and pairwise variants did not transfer to locked/public evaluation [I3][I4][I5]. V18 is materially different because it creates a new observation: the decisive `Dieu/Khoan/Diem` text that the current three-passage view may miss.
4. **The proposal has large parameter safety margin.** V18 reuses all existing models: 330M BQB + 760M CE under the project's conservative accounting + at most 10K selector parameters = **1.09001B**. An optional 0.6B sidecar would bring the total only to **1.69001B**.

The realistic planning range is +0.002 to +0.006 Recall@5 if the evidence audit confirms the hypothesis. This is not a promised gain. The first experiment is deliberately falsifiable and can terminate the direction in one to two days.

> **One-line decision:** invest first in legal-structure evidence packing at the c35 boundary; add a 0.6B reranker or candidate-conditioned QREL-text memory only after V18 proves that it supplies independent signal.

2 Task constraints and the baseline to protect

2.1 Competition objective

The organizer's data overview defines the input as a Vietnamese legal question and the output as relevant document IDs. Macro Recall is primary and Precision is secondary. A query may contain at most five IDs; returning more than five gives both Recall and Precision of zero for that query [I1, pp. 2-3].

Consequences:

- Default to five distinct, valid IDs per query.
- Do not import COLIEE-style adaptive cutoff as a primary idea. Under a recall-first objective and a hard cap of five, returning fewer documents discards recall opportunities even if it improves precision.
- Optimize macro Recall@5. Treat Precision@5 as a safety metric and tie-breaker.

2.2 Current V16b architecture

V16b is the main pipeline at Public LB 0.9210 and locked holdout 0.9228333 [I2].

Layer	Current function	Protection requirement
Stage-1	Legal FTS, phrase/citation, chunk retrieval, dense BQB	Keep initially; coverage is already strong
Safe-tail PRF	Corpus-only PRF with protected top candidates	Keep; it has transferred
c35	Thirty-five candidates passed to reranking	Keep; oracle is 0.967667
CE	Up to 3 passages/doc, max length 768; `best + .15 second + .05 third`	Keep model; audit its evidence view
V16 QREL-memory	Fit-only query-neighbor to document propagation	Keep; it produced real gain
Slot-5 gate	Top four locked; replace tail only with memory and margin safeguards	Keep until new signal transfers

V11h established the Stage-1 to c35 to CE to selector backbone, including 38 query-normalized features and the strict exact-heading rescue threshold of at least 10 [I6]. V16b added fit-only QREL-memory and the slot-5 gate, yielding a net +0.004 on the locked holdout relative to this branch's c35 anchor [I3].

2.3 Strict end-to-end parameter budget

Use the project's conservative counts, not model-card rounding:

Component	Counted parameters
BQB retriever	330,000,000
Vietnamese CE	760,000,000
Tiny selector/gate	<= 10,000
V16b or V18 total	**1,090,010,000 (1.09001B)**

The requirement is strictly less than 4B for the complete pipeline. FP16, INT8, or other quantization does not reduce architectural parameter count. A 4B model cannot be added to the current stack, and should not be treated as compliant through quantization.

3 Bottleneck diagnosis

3.1 The c35 oracle makes this primarily an evidence/reranking problem

On the locked holdout:

$$\text{headroom within c35} = 0.967667 - 0.9228333 = 0.0448337$$

That is much larger than the desired breakthrough increment. P0 further estimated recoverable headroom of 0.0643275 inside c35 versus 0.0315929 outside, so 67.06% of accessible opportunity is already present in the candidate set [I5]. Local experiments agree:

- c50 improved OOF by about 0.002506 but decreased locked performance by 0.002 and scored Public 0.9181 versus 0.9183 for v11h [I4].
- prefix512 improved candidate oracle but produced locked 0.914333 and Public 0.9090 [I4].
- frozen Jina-ColBERT-v2 showed small OOF complement gain but negative locked transfer and poor inference precision [I4].

Candidate expansion has oracle value, but its OOD and selection cost has exceeded its realized gain. It is not the first move.

3.2 Error categories identify missing evidence

The P0 error audit records [I5]:

Error category	Queries	Interpretation
CE-buried	64	Gold is in c35 but CE ranks it too low
Selector-lost	28	Useful signal exists but final selection loses gold
Pruned before c35	50	Candidate work is needed, but this is not the largest tractable group
Unavailable in c200	36	True Stage-1 ceiling; harder and higher risk

The first two groups provide 92 cases that can be attacked without changing candidate generation. V16b also exposes the limit of pure calibration: on locked evaluation it made 177 slot changes, with 9 rescues and 4 destructive changes, for +0.004 net [I3]. V16d improved memory-level Recall@5 and inference precision but left downstream OOF Recall unchanged [I5]. Better memory metrics alone are insufficient if the boundary scorer does not see decisive document evidence.

3.3 The current CE leaves trained context capacity unused

The official `Vietnamese_Reranker` model card describes a BGE reranker v2 m3 derivative trained on approximately 1.1M Vietnamese triplets with maximum sequence length 2304, comprising up to 256 query tokens and 2048 passage tokens [R3]. V16b uses length 768 and at most three passages. Two systematic failures are therefore plausible:

1. The decisive clause is outside the current top three selected passages.
2. The clause text is separated from its title, article, or governing paragraph, so the CE loses scope and applicability.

Simply changing maximum length to 2304 is not the proposal. BReps reports that more raw context can become worse because irrelevant blocks add noise [R5]. KeyB2 finds that key-block preselection beats naive long-input and pooling alternatives [R4]. LawRAG reports that legal chunking around title, article, and paragraph outperforms sequential chunking for Indonesian statutes [R6]. The correct test is **selected, structure-preserving evidence**, not a longer prefix.

4 Research synthesis

4.1 LegalIR competition evidence: scale alone is not the answer

NOWJ@COLIEE 2026 used a four-stage pipeline with candidate filtering, dense retrieval, reranking, and per-query cutoff. For Task 1, the feature-based MLP reranker submission ranked first, while the Qwen generative reranker submission ranked sixth [R1]. This does not prove that MLPs always beat LLM rerankers, but it directly argues against a parameter-scale-first strategy in LegalIR.

The transferable lessons are staged reranking and domain-specific evidence. The adaptive cutoff itself is not transferable because the UIT objective and output cap differ.

SAILER emphasizes that legal documents are long and structured, and that subtle legal elements can determine relevance [R14]. DELTA focuses on key facts and legal discriminativeness rather than semantic similarity alone [R15]. LegalSearchLM reports large gains on its Korean benchmark by representing legal elements explicitly [R16], but its generative design is not a direct fit for this budget or dataset. The useful abstraction is to preserve legal elements and their governing context in the evidence supplied to a compact scorer.

4.2 Long-document reranking: select blocks before interaction

Three recent lines of work converge:

1. **KeyB2** pre-ranks blocks before applying a stronger reranker and reports better effectiveness with lower compute on TREC 2019 DL, Robust04, and MLDR-zh [R4].
2. **BReps with Top-k Interaction Refinement** precomputes block embeddings, selects top blocks, and applies a bounded residual interaction module. It reports NDCG@10 increasing from 0.395 to 0.451 on TREC DL 2023 under the same block budget [R5].

3. **LawRAG** reports that title/article/paragraph segmentation outperforms sequential chunking on Indonesian legal documents [R6].

These results do not guarantee UIT Recall@5 gains. They establish a strong, testable prior: block selection and structure-preserving packing should be audited before replacing the model.

4.3 Lightweight rerankers: a second-stage option, not the first bet

Qwen3-Reranker is available at 0.6B with multilingual, instruction-aware reranking and an Apache 2.0 release [R2]. `jina-reranker-v3.5` is a 0.6B listwise model with cross-document interaction and legal/multilingual training data [R7]. Either fits the parameter budget when added alone:

$$1.09001\text{B} + 0.6\text{B} = 1.69001\text{B} < 4\text{B}$$

However:

- V13 generic CE LoRA and V14 frozen late interaction did not transfer [I4].
- NOWJ's competition evidence does not support assuming the larger/generative reranker will win [R1].
- `jina-reranker-v3.5` is released under a non-commercial license [R7]; competition use must be verified before adoption.
- Qwen3-Reranker-0.6B is a pointwise relevance scorer, not a drop-in equivalent of joint listwise interaction.

A sidecar is valuable only if it produces independent error signal on already improved evidence packs.

4.4 Negative mining: semi-hard beats blindly hardest negatives

Work on Vietnamese legal document retrieval from the SoICT setting compares negative sampling regimes and reports semi-hard negatives outperforming hard and easy negatives. Semi-hard examples were sampled from a retrieved rank band after positives were removed [R8]. This is consistent with the local failure of V13 generic hard-negative LoRA [I4]: the very hardest retrieved items are more likely to contain false negatives or ambiguous legal overlap.

If any later reranker adaptation is attempted:

- use fit-only labels in each fold;
- sample semi-hard rank bands, not the hardest top-n blindly;
- split by law family, template, and near duplicates;
- use low-rank adaptation and early stopping;
- do not rely on an upstream teacher that would make the model lineage exceed the competition cap.

4.5 Memory augmentation: change the memory representation, not just its adapter

MoMA demonstrates that plug-in memories can supplement dense retrievers [R12]. Doc2query shows that questions a document can answer provide useful retrieval representations [R11]. CaseLink and JuriFindIT show that document relations and legislative graphs can help LegalIR [R9][R17].

V16b already has effective fit-only query-neighbor to document propagation [I3]. V16d's flat downstream result warns against adding another adapter over the same representation [I5]. A materially different memory experiment would be **candidate-conditioned QREL-text memory**:

- collect fit-only historical questions associated with each candidate document;
- compare the new query directly with those question prototypes using the existing BQB and CE;
- expose top-1, top-3, log-sum-exp, support count, entropy, and degree-shrunk scores;
- compare each candidate with the current tail anchor;
- remove same-query, near-duplicate, and held-out-family leakage in every fold.

This differs from global V16 propagation and resembles a supervised, cross-fitted doc2query representation. It remains third priority because the V16d result shows that memory signal may fail to cross the final decision boundary.

5 Strategic comparison

Direction	Match to observed bottleneck	Total parameters	Transfer risk	Priority
V18 legal evidence packs + current CE	Creates new evidence for CE-buried and selector-lost cases	**1.09001B**	Low-medium; directly auditable	**1**
0.6B boundary sidecar on evidence packs	Adds an independent scorer and possible listwise interaction	**1.69001B**	Medium; license/OOD	**2**
Candidate-conditioned QREL-text memory	Different from global KNN; exploits fit qrels	**1.09001B**	Medium-high; leakage/collisions	**3**
Legal relation/citation graph	Supported by external LegalIR work	1.09B + tiny head	High given local V16d/V14 evidence	4
c50/c80/prefix expansion	Targets the minority of headroom outside c35	Similar	**Already failed locked/public**	Stop
Generic hard-negative CE LoRA	Changes current CE	Similar	**V13 failed**	Stop
Gate/listwise model over old features	Reorders existing signal	Similar	**P1/P4/V17 failed**	Stop
Qwen 4B/8B reranker	Parameter scale	>4B end-to-end	Non-compliant	Prohibited

5.1 Why V18 is not a repeat of earlier failures

V12 Aspect-MaxSim, V15 pairwise ranking, V17 setwise ranking, and the gate-calibration phases primarily changed aggregation or selection over already observed signals. V18 changes the observation layer:

- reconstruct `Title -> Dieu -> Khoan -> Diem` structure;
- search the full candidate document;
- pack multiple non-redundant legal blocks into the 2304-token window the CE was trained to process;
- score challenger and tail evidence using the same protocol.

If the P0 audit does not find omitted decisive blocks, V18 stops. This is a falsifiable bet rather than an open-ended feature sweep.

6 Proposed architecture: V18 E3B

E3B means **Evidence-Complete Boundary Reranking**. The overall pipeline remains V16b, with a new evidence branch only for candidates close to the output boundary.

6.1 Component A: legal-structure block index

Parse the corpus once, without query labels, using this precedence:

1. Document title and scope metadata.
2. Chapter and section headings.
3. `Dieu` (article).
4. Khoan (clause/paragraph).
5. `Diem` (subclause/item).

Each child block retains its ancestor headings. A `Diem` block must carry labels for the document, article, and clause that govern it. When structured parsing fails, use sentence-complete windows of 160-240 tokens with 32-token overlap, avoiding sentence cuts where possible.

Store offline for each block:

- normalized lexical text;
- exact citation and heading keys;
- BQB embedding;
- BM25/FTS representation;
- token length and structural path.

This adds an index, not a model.

6.2 Component B: multi-signal block proposals

The first implementation runs only on the boundary set:

- V16's top four anchors;
- the current slot-5 tail;
- six c35 challengers with the closest CE, selector, or memory margins.

For every query-document pair, take the union of top blocks from:

1. exact heading/citation matches;
2. BM25 and legal phrase matches;
3. BQB block similarity;
4. the current CE-768 passage score.

Apply a simple MMR or overlap filter so that the pack does not contain four near-duplicate fragments from the same article. Do not train a block selector in the first round.

6.3 Component C: 1024-2304 token evidence packs

Use explicit labels and preserve legal ancestry:

```
[TITLE] document title and scope
[EVIDENCE_1] Article ... Clause ... complete text
[EVIDENCE_2] Article ... Clause ... complete text
[EVIDENCE_3] ...
[CITATION_CONTEXT] ... when present
```

Rules:

- retain V16 query preprocessing;
- select 3-5 non-redundant blocks;
- preserve ancestor headings;
- never append an arbitrary raw prefix without selection evidence;
- test only 1024, 1536, and 2304 token budgets;
- log which block was truncated.

Score each pack once with the current CE. Keep the existing best, second, and third passage scores. New features are pack score, pack-to-tail delta, best-unseen-block delta, structural coverage, citation coverage, and evidence diversity.

6.4 Component D: bounded boundary decision

V18b first uses a frozen, pre-registered decision rule over pack-to-tail deltas and existing V16 safety conditions. Only if the frozen pack provides positive signal does V18c fit a tiny residual model:

- pairwise L2 logistic residual or monotone depth-2 booster;
- input limited to new evidence deltas plus a small set of V16 margin/safety features;
- output permitted to replace slot 5 only;
- V16 memory floor and tail-anchor safeguards retained;
- threshold selected by cross-fitting, never from locked/public feedback.

The small calibrator is not the source of the breakthrough. It is a bounded mechanism for admitting new evidence.

6.5 Component E: unlock slot 4 only after transfer

Slot-5-only protects transfer but caps maximum gain. After V18 transfers once, run a counterfactual slot-risk audit:

- Try a two-slot policy only if replacing slots 4-5 adds at least 0.003 OOF oracle headroom beyond slot-5-only and passes the risk splits.
- Otherwise keep the top four locked.

Do not change evidence and slot policy in the same experiment. A failure would otherwise be uninterpretable.

7 Decision-gated experimental plan

7.1 P0 - V18a evidence audit, 1-2 days, no holdout access

Hypothesis H0: a material share of the 64 CE-buried cases contains a decisive legal block outside the current three passages, and that block can score above the slot-5 anchor.

Setup

- Use clean-complement OOF and fit-only labels.
- Parse legal blocks for the corpus.
- Exhaustively or funnel-score blocks for the 64 CE-buried cases, 28 selector-lost cases, and matched negative controls.
- Measure current top-three block capture, truncation, and structural coverage.
- Do not alter a submission or inspect locked holdout.

Metrics

- `decisive_block_outside_top3_rate`;
- best gold-block rank and margin relative to the tail;
- evidence-pack counterfactual c35 oracle Recall@5;
- parser coverage by document family;
- latency and CE calls/query.

Go/no-go gate

Continue only if all hold:

1. at least 20% of CE-buried cases, or at least 13 of 64, contain a decisive block outside the current top three;
2. evidence-pack counterfactual oracle improves by at least 0.004 Recall@5 over the V16 boundary;
3. ancestor headings are preserved for at least 95% of scored blocks.

If this fails, stop V18 rather than tuning the audit to pass. Move to the candidate-conditioned memory diagnostic.

7.2 P1 - V18b frozen evidence packs, 2-4 days

Hypothesis H1: a structure-selected evidence pack improves candidate-tail discrimination without training a new model.

Use one pre-registered six-cell grid:

- selector A: structure + exact/BM25;
- selector B: exact/BM25/BQB/CE union + MMR;
- context length: 1024, 1536, or 2304;
- no new configuration family after scores are observed.

Evaluate standard OOF, law-family OOF, template OOF, and near-duplicate OOF. Save query-level predictions for paired bootstrap.

Report:

- macro Recall@5 and Precision@5;
- boundary AUPRC and inference precision;
- rescue/destructive ledger;
- performance by error category;
- p50/p95 latency, VRAM, and CE calls/query.

Proceed to P2 only if at least three of four splits have positive Recall@5 delta and either boundary AUPRC increases by at least 0.03 or the frozen rule increases OOF Recall@5 by at least 0.0015.

7.3 P2 - V18c bounded residual gate, 2-3 days

Hypothesis H2: a tiny calibrator restricted to new evidence can turn candidate discrimination into net rescues while preserving V16b's OOD safety.

Setup:

- Cross-fit challenger-versus-tail pairwise examples.
- Permit only two pre-registered learners: L2 logistic residual and monotone depth-2 booster.
- Keep top four locked.
- Keep the V16 memory floor and tail anchor.

Promotion gate before locked holdout

Guardrail	Requirement
Standard clean-complement OOF	>= 0.8880624, or V16 +0.002
Robust composite OOF	>= 0.8855418, or V16 +0.0015
Stress splits	At least 3/4 non-negative; worst > -0.0015
Inference precision	>= 0.87
Paired bootstrap delta	Lower 95% confidence bound > 0
Precision@5	No meaningful decrease

If every guardrail passes, freeze configuration and hashes, then open the locked holdout once.

Locked promotion requirement: Recall@5 >= 0.9248333, no precision loss, and no evidence that a single law/template family creates the whole gain. Only then submit to the Public LB.

7.4 P3 - Optional 0.6B sidecar, only after V18 proves signal

Hypothesis H3: a different reranker architecture adds complementary errors on the same high-quality evidence packs.

Order:

1. Probe frozen `jina-reranker-v3.5` 0.6B listwise over the 10-11 boundary candidates only if its license permits competition use [R7].
2. If license use is unclear or prohibited, probe Apache-2.0 Qwen3-Reranker-0.6B pointwise on each evidence pack [R2].
3. Do not fine-tune the first probe.
4. Measure agreement, unique rescues, destructive changes, and error correlation with the current CE.

Attempt LoRA only if the frozen model adds at least 0.001 unique net OOF recall and has error correlation below 0.85. Use semi-hard rank bands if training proceeds [R8]. The total parameter count with one sidecar is 1.69001B.

7.5 P4 - Candidate-conditioned QREL-text memory

Hypothesis H4: historical questions attached directly to each candidate act like a cross-fitted doc2query representation and separate collisions that global V16 KNN cannot.

Leakage-safe setup:

- Build memory only from each fold's fit partition.
- Exclude exact and near-duplicate query families and shared templates.
- Retrieve the top historical questions attached to each candidate.
- Score query-prototype pairs with existing BQB and/or CE.
- Apply document-degree shrinkage, minimum support, and missingness flags.

Admit these features to V18c only if their boundary oracle adds at least 0.002 OOF after evidence packing and remains positive on three of four stress splits. If only a memory-level metric improves while downstream Recall remains flat, stop immediately, following the V16d lesson.

7.6 P5 - Optional two-slot policy

Run only after V18 has passed locked evaluation. Add no new model or evidence feature in this phase. Unlock slots 4-5 only if:

- counterfactual oracle adds at least 0.003 OOF beyond slot-5-only;
- expected slot-4 inference precision is at least 0.90;
- paired-bootstrap lower confidence bound is positive on the robust complement;
- no stress split falls by more than 0.001.

8 Evaluation discipline and OOD controls

8.1 Use the correct anchor

Do not compare V18 to the 0.901866 OOF value from the V11h lineage because its data and anchor differ. The correct V16 clean-complement anchor is **0.8860623782**, with robust score **0.8840418489** [I5]. Every result table should record lineage, split hash, candidate cache hash, and feature cache hash.

8.2 Minimum report for every run

1. Macro Recall@5 and Precision@5.
2. Query-level delta from V16b.
3. Standard, law-family, template, and near-duplicate splits.
4. Paired bootstrap with 10,000 samples and 95% interval.
5. Rescue/destructive ledger and inference precision.
6. CE-buried, selector-lost, pruned, and unavailable buckets.
7. p50/p95 latency, peak VRAM, and CE calls/query.
8. Parameter ledger for every inference model and any model counted in the submission lineage.

8.3 Locked and Public policy

- P0-P2 do not inspect locked holdout.
- Select one frozen candidate by the pre-registered robust rule.
- Open locked once for promotion.
- Use Public LB as final confirmation, not hyperparameter feedback.
- Reject negative locked delta or a clear precision loss; do not make a holdout-informed micro-fix.

9 Expected impact, cost, and risks

9.1 Planning estimates

No paper supports a direct conversion from NDCG or MRR gains to UIT macro Recall@5. These are planning priors only:

Phase	Expected delta if its hypothesis is true	Confidence
V18 evidence audit/frozen pack	+0.0015 to +0.004 OOF	Medium-high
Bounded residual	+0.0005 to +0.002 additional	Medium
0.6B sidecar	+0.0005 to +0.002 additional	Low-medium
QREL-text memory	+0.0005 to +0.0015 additional	Low-medium

Do not add these ranges mechanically because rescues will overlap. A successful cycle could plausibly reach locked 0.925-0.929, but the promotion gate asks only for +0.002 to avoid discarding a small real gain.

9.2 Principal risks and mitigations

Risk	Detection	Mitigation
Longer pack adds noise	2304 loses to 1536; destructive rate rises	MMR, heading retention, selected blocks instead of prefix
Structural parser errors	Low coverage for certain law families	Sentence-complete fallback; 95% audit gate
Gate overfit	Standard rises while family/template falls	Robust selection and paired bootstrap
Memory leakage	Near-duplicate split rises abnormally	Fit-only build, family exclusion, hash audit
Sidecar license	Non-commercial restriction	License check; Apache Qwen fallback
Compute inflation	p95 or VRAM exceeds deployment budget	Boundary-only scoring, precomputed block index, funnel
Parameter violation	A model or teacher is omitted from ledger	Conservative full-lineage count; no 4B model/teacher

10 Execution sequence

10.1 First seven days

1. Freeze V16b outputs, split hashes, and error-bucket IDs.
2. Implement legal parser and unit tests for title, article, clause, and item structure.
3. Build the block index and token/truncation dashboard.
4. Run V18a on 64 CE-buried + 28 selector-lost cases and matched controls.
5. Make the P0 go/no-go decision without opening holdout.

10.2 Second week

1. If P0 passes, run the six frozen evidence-pack configurations.
2. Select exactly one configuration by robust composite.
3. Run the bounded residual cross-fit.
4. Complete bootstrap, rescue ledger, compute, and parameter audits.
5. Open locked only if every P2 guardrail passes.

10.3 After the first promoted candidate

1. If locked is at least 0.9248333, submit Public and freeze V18.
2. Audit the slot 4-5 oracle without changing the policy.

3. If gain remains insufficient and evidence signal is strong, probe one 0.6B sidecar.
4. Only then test candidate-conditioned QREL-text memory.

11 Explicit stop list

- Do not continue c50/c80/prefix expansion without a genuinely new retriever signal.
- Do not fine-tune the CE with blindly hardest negatives.
- Do not add another selector/listwise model over the unchanged V16 feature set.
- Do not switch to a raw 2304-token prefix; evidence selection is essential.
- Do not import COLIEE adaptive cutoff into a recall-first, max-five task.
- Do not add Qwen 4B; the end-to-end pipeline would be non-compliant.
- Do not use Public or locked performance to choose hyperparameters.
- Do not call a memory experiment successful when only its internal metric improves.

12 Evidence-gap matrix

Decision claim	Supporting evidence	Contradiction or gap	Confidence
The main bottleneck is inside c35	+0.044834 c35 oracle headroom; 67.06% inside	Error buckets depend on audit definitions	High
Evidence packing is the best first bet	CE uses 768/3 passages; trained at 2304; KeyB2/BReps/LawRAG	No UIT P0 audit yet	Medium-high
A larger model is not required	NOWJ MLP ranked first; Qwen submission sixth; V13/V14 failed	Different datasets and tasks	Medium
A 0.6B sidecar may add signal	Qwen/Jina benchmarks and cross-document design	License and OOD transfer unknown	Low-medium
QREL-text memory differs from V16	Candidate-conditioned/doc2query rationale	V16d improved memory metric but not downstream	Low-medium
Candidate expansion is not first priority	c50/prefix/public negatives; minority headroom outside	32.94% of headroom remains outside	High

The most important unknown is the share of the 64 CE-buried cases whose decisive block is outside the current top three passages. V18a has the highest information value because it can invalidate the complete strategy in one to two days.

13 Claim-to-source ledger

Claim	Primary source	Use in the decision
UIT max five, >5 yields zero, Recall primary	I1	Output policy; rejects adaptive cutoff
V16b 0.9210/0.9228333 and 1.09001B budget	I2-I3	Baseline and parameter audit
c35 oracle, headroom, error categories	I4-I5	Bottleneck diagnosis
Current CE trained at length 2304	R3	Context-capacity audit
Selected key blocks beat naive long input	R4-R6	Evidence-pack architecture
LegalIR MLP can beat generative reranker	R1	Counterweight to model-scale bias
0.6B reranker alternatives	R2, R7	Optional sidecar

Claim	Primary source	Use in the decision
Semi-hard negatives in Vietnamese LegalIR	R8	Adaptation protocol, if needed
Query prototypes and legal graphs	R9-R12, R17	Third direction, not a settled conclusion

14 Internal project sources

- **[I1]** `DSC2026_Task1_LegalIR_Data_Overview(1).pdf`, UIT DSC 2026 organizer data overview, pp. 2-3.
- **[I2]** `CURRENT_STATE(1).md`, current project state and V16b baseline.
- **[I3]** `V16_RGM_DISCLOSURE(1).md`, QREL-memory, features, and slot-5 gate.
- **[I4]** `EXPERIMENT_LOG(1).md`, c50, prefix512, V12-V15, and negative branches.
- **[I5]** `V16_BOUNDARY_RESEARCH_DISCLOSURE.md`, P0-P4, error taxonomy, OOD/bootstrap, and stop rules.
- **[I6]** `FINAL_V11H_END_TO_END_SUMMARY(2).md`, Stage-1 to c35 to CE to selector.
- **[I7]** `README(1).md` and `README(2).md`, execution, submission, and model-budget constraints.

15 Web and academic sources

- **[R1]** Ngo et al. (2026), NOWJ@COLIEE 2026: Adaptive Pipelines for Legal Retrieval and Reasoning, arXiv. https://arxiv.org/abs/2607.16603
- **[R2]** Zhang et al. (2025), Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models, arXiv. https://arxiv.org/abs/2506.05176
- **[R3]** thanhtantran, Vietnamese_Reranker model card, Hugging Face. https://huggingface.co/thanhtantran/Vietnamese_Reranker
- **[R4]** Li et al. (2024/2025), KeyB2: Selecting Key Blocks is Also Important for Long Document Ranking with Large Language Models, arXiv. https://arxiv.org/abs/2411.06254
- **[R5]** Li, Gaussier, and Zhou (2025/2026), Efficient Long-Document Reranking via Block-Level Embeddings and Top-k Interaction Refinement, arXiv. https://arxiv.org/abs/2501.17039
- **[R6]** Lai et al. (2026), LawRAG: Indonesian legal document retrieval-augmented generation with specialized chunking and reranking strategies, Data Technologies and Applications. https://doi.org/10.1108/DTA-03-2025-0195
- **[R7]** Nasika et al. (2026), jina-reranker-v3.5: An Efficient Listwise Reranker with Hybrid Attention and Self-Distillation, arXiv. https://arxiv.org/abs/2607.18152
- **[R8]** Le et al. (2025), Optimizing Legal Document Retrieval in Vietnamese with Semi-Hard Negative Mining, arXiv. https://arxiv.org/abs/2507.14619
- **[R9]** (2025), CaseLink: Inductive Graph Learning for Legal Case Retrieval, arXiv. https://arxiv.org/abs/2505.20743
- **[R10]** Shao et al. (2020), BERT-PLI: Modeling Paragraph-Level Interactions for Legal Case Retrieval, IJCAI. https://www.ijcai.org/proceedings/2020/484
- **[R11]** Nogueira et al. (2019), Document Expansion by Query Prediction (doc2query), arXiv. https://arxiv.org/abs/1904.08375
- **[R12]** Microsoft Research (2023), Augmenting Zero-Shot Dense Retrievers with Plug-in Mixture-of-Memories (MoMA). https://www.microsoft.com/en-us/research/publication/augmenting-zero-shot-dense-retrievers-with-plug-in-mixture-of-memories/
- **[R13]** Wang et al. (2025), jina-reranker-v3: Last but Not Late Interaction for Document Reranking, arXiv. https://arxiv.org/abs/2509.25085
- **[R14]** Li et al. (2023), SAILER: Structure-aware Pre-trained Language Model for Legal Case Retrieval, arXiv. https://arxiv.org/abs/2304.11370
- **[R15]** (2025), DELTA: A Discriminative Legal Retrieval Model with Key Facts and Legal Elements, AAAI Digital Library. https://ojs.aaai.org/index.php/AAAI/article/view/34914
- **[R16]** Kim, Lee, and Hwang (2025), LegalSearchLM: Rethinking Legal Case Retrieval as Legal Elements Generation, EMNLP, ACL Anthology. https://aclanthology.org/2025.emnlp-main.225/
- **[R17]** Dalla Noce et al. (2026), JuriFindIT: an Italian legal retrieval dataset, Findings of EACL, ACL Anthology. https://aclanthology.org/2026.findings-eacl.221/

16 Limitations

- Some 2026 sources are recent preprints and may not yet have complete peer review. They are used to form hypotheses, not to override local OOF evidence.
- NDCG, MRR, and F1 results from TREC, COLIEE, and SoICT cannot be converted directly into UIT macro Recall@5.
- This report did not access hidden test labels and does not use Public LB iteration as evidence.
- Gain estimates are planning priors. The P0-P2 gates determine whether implementation should continue.

Final recommendation: start with the V18a evidence audit. If decisive-block rate and oracle gain fail the gate, stop in one to two days. If they pass, V18 E3B is the only current direction that simultaneously creates new signal, preserves V16b's OOD discipline, and keeps the full pipeline at 1.09001B parameters.